Homework Assignment 4

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Midterm 1

Chapter 1:

- $\mathbb{N} \iff$ Well Ordering Principle $\iff$ Mathematical Induction
- A is countable $\iff A \sim \mathbb{N}$.
- Axiom of Completeness $\iff$ Any non-empty $A \subset \mathbb{R}$ bounded above has $\sup A$.
- Nested Interval Property:

$$[a_1, b_1] \subseteq [a_2, b_2] \subseteq \cdots \subseteq [a_n, b_n] \implies \bigcap_{n=1}^{\infty} [a_n, b_n] \neq \varnothing.$$

- $\mathbb{R}$ and $\mathbb{R} \setminus \mathbb{Q}$ is not countable.

Chapter 2:

- Convergent Sequences:

$$a_n \to a \iff \forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } n \geq N \implies |a_n - a| < \epsilon.$$

- Monotone Convergence Theorem: a_n is bounded and monotonic $\implies a_n$ converges.
- Bolzano Weierstrass: Every bounded sequence contains a convergent subsequence.
- Cauchy Sequences:

$$a_n \text{ is cauchy} \iff \forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } n, m \geq N \implies |a_n - a_m| < \epsilon$$

- Cauchy Criterion: Convergent $\iff$ Cauchy.

Practice Midterm 1 5. We say that a_n diverges to ∞ if for every $M > 0$ there exists $N \in \mathbb{N}$ such that $n \geq N \implies a_n > M$. Suppose $a_n \to \infty$ and $b_n \to b > 0$. We want to show that

$$a_n b_n \to \infty.$$

Let $M > 0$. Since b_n converges, choose $N_1 \in \mathbb{N}$ such that

$$n \geq N_1 \implies |b_n - b| < \frac{b}{2}.$$

So $b_n > \frac{b}{2}$ for every $n \geq N_1$. Since a_n diverges, choose $N_2 \in \mathbb{N}$ such that

$$n \geq N_2 \implies a_n > \frac{2M}{b}.$$

Set $N = \max\{N_1, N_2\}$. For every $n \geq N$ we have

$$a_n b_n > \frac{2M}{b} \cdot \frac{b}{2} = M.$$

Therefore $a_n b_n \to \infty$.